

CS621/CSL622

Quantum Computing For Computer Scientists

Quantum Circuits and Protocols

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- Time goes from left to right.
- Horizontal lines represent qubits.
- Operations and measurements are represented by various symbols.
- Hadamard transform applied to a single qubit



- Double lines indicate classical bits



Example

Consider two qubits X and Y in the superposition

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

- Using the Dirac Notation:

$$\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

Operation: $H \otimes I$

A Hadamard transform to the first qubit and nothing to the second qubit

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$$

- In Dirac notation: $\frac{1}{2} |00\rangle + \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle$
- Can perform this computation directly with the Dirac notation?

Note

$$H|0\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \quad H|1\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

- Starting superposition:

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle = \frac{1}{\sqrt{2}}|0\rangle|0\rangle + \frac{1}{\sqrt{2}}|1\rangle|1\rangle$$

- Superposition after $H \otimes I$

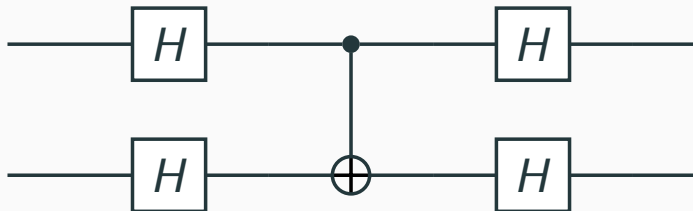
$$\frac{1}{\sqrt{2}}(H|0\rangle)|0\rangle + \frac{1}{\sqrt{2}}(H|1\rangle)|1\rangle$$

- Using values of $H|0\rangle$ and $H|1\rangle$

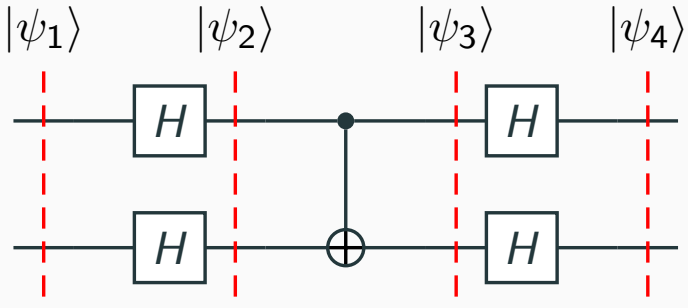
$$\begin{aligned} & \frac{1}{\sqrt{2}}(H|0\rangle)|0\rangle + \frac{1}{\sqrt{2}}(H|1\rangle)|1\rangle \\ &= \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \right) |0\rangle + \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \right) |1\rangle \\ &= \frac{1}{2}|0\rangle|0\rangle + \frac{1}{2}|0\rangle|1\rangle + \frac{1}{2}|1\rangle|0\rangle - \frac{1}{2}|1\rangle|1\rangle \\ &= \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle \end{aligned}$$

Verdict

Dirac Notation can be directly employed to handle operations in quantum circuits

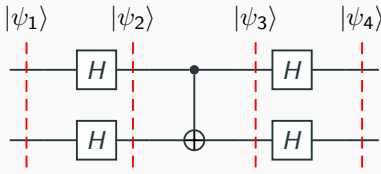


- What does this circuit do?



Case-1: $|\psi_1\rangle = |00\rangle$

Circuit Slicing



$$|\psi_2\rangle = \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right)$$

$$= \frac{1}{2} |00\rangle + \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle + \frac{1}{2} |11\rangle$$

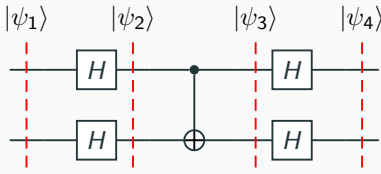
$$|\psi_3\rangle = \frac{1}{2} |00\rangle + \frac{1}{2} |01\rangle + \frac{1}{2} |11\rangle + \frac{1}{2} |10\rangle$$

$$= \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right)$$

$$|\psi_4\rangle = |00\rangle$$

Case-2: $|\psi_1\rangle = |01\rangle$

Circuit Slicing



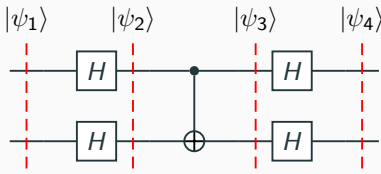
$$|\psi_2\rangle = \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) \left(\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \right)$$

$$= \frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle$$

$$|\psi_3\rangle = \frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle + \frac{1}{2} |11\rangle - \frac{1}{2} |10\rangle$$

$$= \left(\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \right) \left(\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \right)$$

$$|\psi_4\rangle = |11\rangle$$



$$|\psi_2\rangle = \left(\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \right) \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right)$$

$$= \frac{1}{2} |00\rangle + \frac{1}{2} |01\rangle - \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle$$

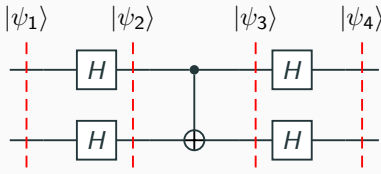
$$|\psi_3\rangle = \frac{1}{2} |00\rangle + \frac{1}{2} |01\rangle - \frac{1}{2} |11\rangle - \frac{1}{2} |10\rangle$$

$$= \left(\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \right) \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right)$$

$$|\psi_4\rangle = |10\rangle$$

Case-4: $|\psi_1\rangle = |11\rangle$

Circuit Slicing



$$|\psi_2\rangle = \left(\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \right) \left(\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \right)$$

$$= \frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle - \frac{1}{2} |10\rangle + \frac{1}{2} |11\rangle$$

$$|\psi_3\rangle = \frac{1}{2} |00\rangle - \frac{1}{2} |01\rangle - \frac{1}{2} |11\rangle + \frac{1}{2} |10\rangle$$

$$= \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) \left(\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \right)$$

$$|\psi_4\rangle = |01\rangle$$

Input	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 11\rangle$
$ 10\rangle$	$ 10\rangle$
$ 11\rangle$	$ 01\rangle$



- The roles of **control** and **target** effectively **switched** in the CNOT gate because of the Hadamard transforms

1. Quantum Computing for Computer Scientists, by Noson S. Yanofsky, Mirco A. Mannucci
2. Quantum Computing Explained, David McMahon. John Wiley & Sons
3. Lecture Notes on Quantum Computation, John Watrous, University of Calgary
 - <https://cs.uwaterloo.ca/~watrous/QC-notes/>